CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors. Follow this format for each person.

*NAME: Shoele, Kourosh

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0003-2810-0065

*POSITION TITLE: Associate Professor

*ORGANIZATION AND LOCATION: Florida State University, Tallahassee, Florida, United States

Proposals and Active Projects

***Proposal/Active Project Title:** Collaborative Research: FIRE-MODEL: Advancing Wildland Fire Modeling by Capturing Unresolved Canopy Dynamics

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** National Science Foundation

***Primary Place of Performance:** Florida State University

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2026

***Proposal/Active Project End Date: (MM/YYYY):** 01/2029

***Total Anticipated Proposal/Project Amount:** $606,882

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.8
2027	0.8
2028	0.8

***Overall Objectives:**

The research goal of this proposal is to address the following fundamental question related to advancing our modeling capabilities and understanding of understory fire dynamics under the effect of canopies.

***Statement of Potential Overlap:**

NA

***Proposal/Active Project Title:** Locomotion Dynamics in Yield Stress Fluids

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** National Science Foundation

***Primary Place of Performance:** Florida State University

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2025

***Proposal/Active Project End Date: (MM/YYYY):** 08/2028

***Total Anticipated Proposal/Project Amount:** $500,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.5
2028	0.5

***Overall Objectives:**

The main objective of this proposal is to assess the dynamics of a self-propelled swimmer in a yield stress fluid.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:** Identifying the Regimes and Acoustics of Cryogenic Boiling heat Transfer for Accelerator Applications

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** Department of Energy

***Primary Place of Performance:** Florida A&M University

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2025

***Proposal/Active Project End Date: (MM/YYYY):** 12/2027

***Total Anticipated Proposal/Project Amount:** $797,933

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.75

Year	Person Months
2026	0.75
2027	0.75

***Overall Objectives:**

The main objective of this research is to quantify how to integrate the boiling heat transfer in the design of linear cool copper technology accelerators

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:** Quantum-based Homogenized Multiscale Simulation of Turbulent Flows

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** Air Force Office of Scientific Research

***Primary Place of Performance:** Florida A&M University

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2025

***Proposal/Active Project End Date: (MM/YYYY):** 12/2027

***Total Anticipated Proposal/Project Amount:** $713,842

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	1
2026	1
2027	1

***Overall Objectives:**

This research explores proper turbulent formulation for the use of quantum-computing to accelerate flow dynamics simulations

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:**	Real-time Gust Prediction with Concurrent Use of Fast Aerodynamic Modeling and Machine Learning Techniques
***Status of Support:**	Current
Proposal/Award Number:	
***Source of Support:**	Air Force Office of Scientific Research
***Primary Place of Performance:**	FAMU-FSU College of Engineering
***Proposal/Active Project Start Date: (MM/YYYY):**	01/2025
***Proposal/Active Project End Date: (MM/YYYY):**	12/2027
***Total Anticipated Proposal/Project Amount:**	$629,787

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	1
2026	1
2027	1

***Overall Objectives:**

To use machine learning and reduced order models to identify incoming gusts using surface sensors.

***Statement of Potential Overlap:**

NA

***Proposal/Active Project Title:**	A Comprehensive Investigation of Nonlinear Shock-Induced Flutter in High-Speed Flows
***Status of Support:**	Current
Proposal/Award Number:	
***Source of Support:**	National Science Foundation
***Primary Place of Performance:**	FAMU-FSU College of Engineering
***Proposal/Active Project Start Date: (MM/YYYY):**	12/2023
***Proposal/Active Project End Date: (MM/YYYY):**	12/2026
***Total Anticipated Proposal/Project Amount:**	$524,764

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2024	0.75
2025	0.75
2026	0.75

***Overall Objectives:**

Exploring new types of fluttering modes in high-speed flows

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:** CAREER: Multiscale Turbulent Flow Interaction with Flexible Branched Trees for Storm Impact Research

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** National Science Foundation

***Primary Place of Performance:** FAMU-FSU College of Engineering

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2020

***Proposal/Active Project End Date: (MM/YYYY):** 12/2024

***Total Anticipated Proposal/Project Amount:** $500,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2020	1
2021	1
2022	1
2023	1
2024	2

***Overall Objectives:**

To understand the turbulent boundary layer interaction with canopy of branched trees and finding the leading cause of large scale turbulent coherent structures in TBL over vegetated canopies.

***Statement of Potential Overlap:**

none

***Proposal/Active Project Title:** Collaborative Research:SCH: AI-powered and Physics-Infused Ultrasound for Accessible Cardiac Diagnostics

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** National Science Foundation

***Primary Place of Performance:** Florida State University

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2026

***Proposal/Active Project End Date: (MM/YYYY):** 08/2030

***Total Anticipated Proposal/Project Amount:** $422,531

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	1
2028	1
2029	1
2030	1

***Overall Objectives:**

The proposal will develop a novel AI-enabled and physics-constrained framework for reconstructing high-resolution 3D and time-resolved (4D) cardiac function from sparse and noisy 2D ultrasound.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:** Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** DOE

***Primary Place of Performance:** Florida State University

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2026

***Proposal/Active Project End Date: (MM/YYYY):** 03/2027

***Total Anticipated Proposal/Project Amount:** $296,096

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1

***Overall Objectives:**

(this Proposal) Develop mass and volume preserving data assimilation technique for multiphase flow in order to develop AI generated sub-scale models.

***Statement of Potential Overlap:**

NA (this Proposal)

<u>Addendum</u>

Do the Notice of Funding Opportunity (NOFO) or Award instructions require a Current and Pending (Other) Support addendum?

No

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Current and Pending (Other) Support Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

If this is an R&D proposal/award, I also certify to the following statement:

Within the past 12 months I have completed research security training meeting the requirements in SEC. 10634(b) of 42 USC 19234.

Certified by Shoele, Kourosh in SciENcv on 2026-04-22 23:01:17